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Studying the design parameters of an auger filler in packaging black pepper seeds using the discrete element method

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ABSTRACT

Auger fillers are vastly used in the food industries for packaging edible seeds, spices and powders. This research studies the effects of design parameters on the mass flow rate of black pepper seeds discharged from an auger-filler and the required input power. These parameters were auger angular velocity, diameter and pitch-to-diameter ratio and agitator angular velocity. It is intended to reach an optimal design of the auger filler with higher efficiency. To investigate the effects of the parameters and to reduce the number of simulations, the Taguchi algorithm is used. The obtained data from 16 discrete element simulations are analyzed using analysis of variance (ANOVA). It is concluded that the most effective parameters on the output mass flow rate and the required power are the auger outer diameter and pitch-to-diameter ratio, respectively. Finally, the best outcome corresponding to the maximum mass flow rate and the minimum required power is found to occur at the auger angular velocity of 550 rpm, the agitator angular velocity of 25 rpm, the auger outer diameter of 2 inches, and pitch-to-diameter ratio of 1.5.

KEYWORDS

Auger filler; discrete element modeling; food packaging; granular flow; Taguchi's method

1. Introduction

Nowadays, food waste is one of the main problems all over the world, not only causing economic and industrial costs, but also environmental issues. By increasing the population and demands for food, this problem has become more critical (Racionero et al. 2018). Smart material handling techniques (Ghorbanirezaei et al. 2019) in food packaging can be considered as a suitable solution to reduce food waste. Auger filler is one of the machines used in food industries for the intelligent packaging of grains such as pepper, coffee and corn seeds. The main parts of this device are motor, in feed duct, hopper tank, agitator, auger, funnel (outlet pipe) and spinner plate located in the funnel. A schematic view of a semi-automatic auger filler is illustrated in Figure 1.

The auger filler can package powder grains, granular and in some cases liquids. In this research, packaging black pepper seeds, as an example of granular packaging, is simulated in a semi-automatic auger filler of All-fill Company (www.all-fill.com).

Augers are also used in screw conveyors to translate grains, seeds, cement powders, soil and sand. Owen and Cleary (2009) investigated the effect of screw angular velocity, its angle of inclination and shape of particles on mass flow rate and power consumption. The results showed that the mass flow rate increased by increasing screw angular velocity and decreased by increasing the angle of inclination. However, the required power increased by increasing both screw angular velocity and angle of inclination. The shape of particles had no significant influence on mass flow rate,

while was highly effective on the power consumption. Spherical particles needed less power to be translated than sharp-edged particles. Hou, Dong, and Yu (2014) studied the effect of the screw angular velocity and particles cohesive force on mass flow rate in a screw conveyor. Similar to the previous research, they observed an increase in the mass flow rate by increasing the screw angular velocity. However, cohesive force between particles had a negative effect on the mass flow rate. Wang et al. (2019), conducted DEM simulations to investigate the particles flow behavior affected by the screw angular velocity and the surrounding pipe filling level inside a screw conveyor. They concluded that the mass flow rate increased by increasing both the screw angular velocity and the pipe filling level. Furthermore, particles temperature was observed to decrease by increasing the media porosity. Pezo et al. (2015) investigated the effect of different types of auger on the particles mixing quality. They observed higher mixing quality of particles in the case of single flight screw compared to other studied cases.

One of the methods used for modeling and analysis of grains motion is a discrete element method (DEM). This method has been vastly used to model different grains motion in various food industries. For instance, the DEM method was performed to improve grinding black pepper seeds, where it was concluded that cooling the seeds increases the resulted spice quality (Ghodki and Goswami 2017). This method has also been used to study the stresses and seeds breakage phenomenon occurred in processes such as conveying or grinding of seeds (Ghodki, Kumar, and Goswami 2018). It was observed that 50 seeds might be

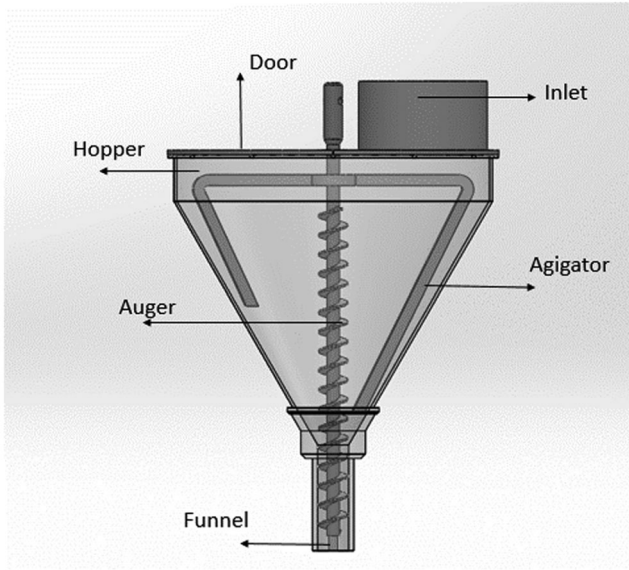


Figure 1. Schematic of an auger filler depicting its main parts.

broken into 250 particles after grinding (Ghodki, Kumar, and Goswami 2018). The discrete elements method is utilized to calculate the critical height of rice grains in silos. The particles begin to discharge a silo by reaching the critical height, which depends on several factors such as the silo walls material and its hopper cone angle (Zhang et al. 2018). DEM simulations were performed to obtain the stress distribution on a silo wall, showing that the wall shear stress increases as particles descend the silo, and the normal stress is higher for larger particles (Horabik, Parafiniuk, and Molenda 2018). In another DEM study, it was found that being transported by trucks, apple intactness depends on their acceleration and vibration frequency (Zeebroeck et al. 2006). The combined computational fluid dynamics (CFD) and DEM, also called CFD-DEM, is broadly used in processes like drying or mixing of grains in food industries (Azmir, Hou, and Yu 2020). By conducting DEM simulations, grains flow in auger fillers, and different types of mixers such as V-mixers was analyzed to improve the machines design (Kuo et al. 2002). Gomes de Freitas et al. (2020) performed the experimental characterization of a novel industrial bulk solids feeder. It was shown that their developed device has almost similar transport rate and gas consumption to the full-size conventional type. The results confirmed the initial hypothesis of lower energy consumption (Gomes de Freitas et al. 2021).

In this study, DEM simulations are performed to analyze the flow of black pepper seeds in a semi-automatic auger filler. The effects of auger angular velocity, diameter and pitch to diameter ratio (P/D), and agitator angular velocity on seeds mass flow rate at the filler outlet and the motor required power is studied. These four parameters have not been studied before simultaneously to achieve an optimum condition in the design of auger fillers. This aim is reached by analysis of variance method applied to the DEM simulations results. These are important parameters determining the seeds motion due to variations of the contact forces

between the seeds and machine parts. Finally, to reach a better design of the auger filler, an optimized condition is proposed to have maximum seeds mass flow rate by consuming minimum input power.

2. Theory

2.1. Soft sphere impact model

The theory of soft sphere collision is used in this research. The equations of motion of N particles in a system are derived from Newton's second law as follows (Norouzi et al. 2016):

$$m_i \frac{dv_i}{dt} = f_i(x_j, v_j, \phi_j, \omega_j) \text{ for } j = 1, 2, \dots, N \quad (1a)$$

$$I_i \frac{d\omega_i}{dt} = M_i(x_j, v_j, \phi_j, \omega_j) \text{ for } j = 1, 2, \dots, N \quad (1b)$$

The quantities in the Equations (1a) and (1b) are: m_i : Mass of the particle i ; I_i : Mass moment of inertia of particle i ; x_j : Cartesian location of Particle j ; ϕ_j : The angular position of particle j ; v_i : The translational velocity of particle i ; ω_j : The angular velocity of particle j .

A schematic view of two colliding particles is shown in Figure 2.

The resultant contact force between the two particles is (Norouzi et al. 2016):

$$f_{ij} = f_{ij}^n e_n + f_{ij}^t e_t, \quad (2)$$

where f_{ij}^n is the force acting in the normal direction to the tangent plane of two particles at their contact point and f_{ij}^t is the tangential force between the particles.

Soft particles collision can be considered to follow a viscoelastic model consisting of a nonlinear spring and a linear damper between the particles in the normal and tangential directions (Figure 3) (Norouzi et al. 2016). The normal and tangential forces between two particles, f_{ij}^n and f_{ij}^t are obtained from the following equations:

$$f_{ij}^n = -K_n(\delta_n)_{ij}^{\frac{3}{2}} - C_n(v_n)_{ij} \quad (3a)$$

$$f_{ij}^t = -K_t(\delta_t)_{ij}^{\frac{3}{2}} - C_t(v_t)_{ij}. \quad (3b)$$

Equations (3a) and (3b) are constructed based on Hertz contact theory, where K and C are the effective stiffness and damping of particles and δ and v are overlaps and, relative velocities of particles at their contact point, respectively. Superscripts n and t denote the normal and tangential directions, respectively. If the tangential force between two particles exceeds the maximum possible friction force being $\mu_s f_{ij}^n$, the frictional element works and slip occurs, such that the tangential force reaches the maximum possible friction (Figure 3(b)) (Norouzi et al. 2016). Otherwise, the tangential force calculated from Equations (3b) should be considered.

Having the contact stiffness and damping calculated, the elastic and dissipative forces in the normal direction are obtained from the following equations (Norouzi et al. 2016):

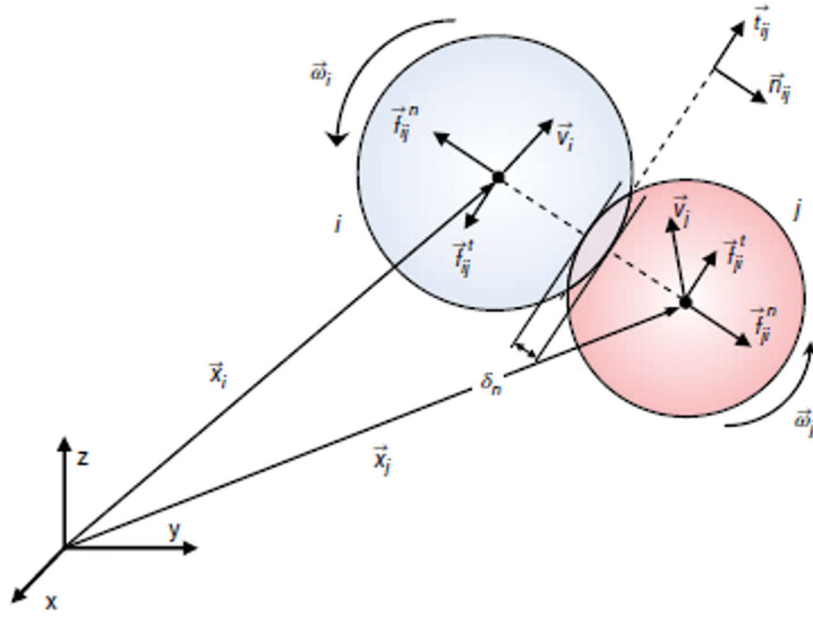


Figure 2. Schematic of two particles colliding each other and the essential quantities (Norouzi et al. 2016).

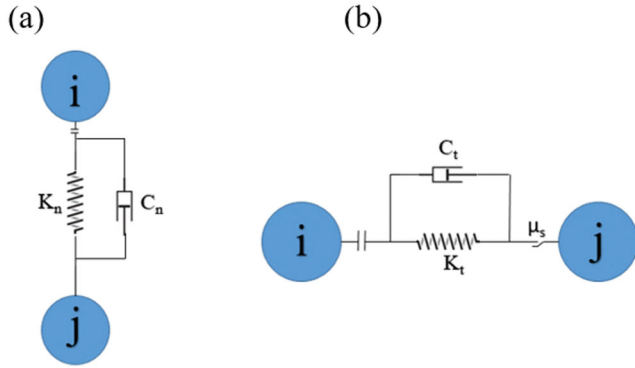


Figure 3. Viscoelastic model of impact force between two colliding particles in (a) normal and (b) tangential directions (Norouzi et al. 2016).

$$(f_{ij}^n)^e = \frac{4}{3} \overbrace{E_{eff} (R_{eff})^{1/2}}^{K_n} (\delta_n)_{ij}^{\frac{3}{2}} \quad (4a)$$

$$\left(f_{ij}^n \right)^d = \left(\overbrace{2 \sqrt{\frac{5}{6}} \frac{\ln(e)}{\sqrt{\ln^2(e) + \pi^2}} \sqrt{8 m_{eff} G_{eff} \sqrt{R_{eff} (\delta_n)_{ij}}}}^{C_n} \right) (v_n)_{ij}. \quad (4b)$$

In Equations (4a) and (4b), e is the coefficient of restitution measured in the drop test as the ratio of post-impact to pre-impact relative particles velocities. m_{eff} , R_{eff} , G_{eff} and E_{eff} are effective mass, effective radius, effective shear modulus and effective Young's modulus, respectively, and are calculated through Equations 5(a)–(5d):

$$\frac{1}{m_{eff}} = \frac{1}{m_i} + \frac{1}{m_j} \quad (5a)$$

$$\frac{1}{R_{eff}} = \frac{1}{R_i} + \frac{1}{R_j} \quad (5b)$$

$$\frac{1}{G_{eff}} = \frac{2(2 - \nu_i)(1 + \nu_j)}{E_i} + \frac{2(2 - \nu_j)(1 + \nu_i)}{E_j} \quad (5c)$$

$$\frac{1}{E_{eff}} = \frac{1 - \nu_i^2}{E_i} + \frac{1 - \nu_j^2}{E_j}. \quad (5d)$$

Parameters R , E and ν are radius, Young's modulus, and Poisson's ratio of particles, respectively.

The elastic and dissipative forces in the tangential direction are calculated by Equations (6a) and (6b), respectively.

$$(f_{ij}^t)^e = \overbrace{8 G_{eff} (R_{eff} \delta_n)^{1/2}}^{K_t} (\delta_t)_{ij} \quad (6a)$$

$$\left(f_{ij}^t \right)^d = \left(\overbrace{2 \sqrt{\frac{5}{6}} \frac{\ln(e)}{\sqrt{\ln^2(e) + \pi^2}} \sqrt{8 m_{eff} G_{eff} \sqrt{R_{eff} (\delta_n)_{ij}}}}^{C_t} \right) (v_t)_{ij}. \quad (6b)$$

The resultant moment, M_{ij} , applied by the particles to each other is obtained from the following equation:

$$M_{ij} = M_{ij}^t + M_{ij}^r. \quad (7)$$

M_{ij}^t is the torque produced by the tangential force and is responsible for the particles rotation, while M_{ij}^r , being the rolling resistance moment, is due to the elastic deformations at the particles surface and prevents the particles from rotating. These two moments are defined in Equation (8) in which e_ω is the unit vector in the direction of particles relative angular velocity.

$$\mathbf{M}_{ij}^t = R_i \mathbf{e}_n \times \mathbf{f}_{ij} \quad (8a)$$

$$\mathbf{M}_{ij}^r = -\mu_r R_{eff} f_{ij}^n \left(\frac{\omega_i - \omega_j}{|\omega_i - \omega_j|} \right) \mathbf{e}_\omega \quad (8b)$$

Choosing a suitable time step is very important since, an incorrectly calculated time step may result in instability or large time consumption of the simulation. The time step should be 20% to 40% of the critical time step calculated from Rayleigh's equation (Li, Xu, and Thornton 2005):

$$\Delta t_{crit} = \frac{\pi R_p}{0.1631 v + 0.8766} \sqrt{\frac{\rho}{G}} \quad (9)$$

In Equation (9), Δt_{crit} , R_p , ρ and G and v are the critical time step, radius of particles, density and shear modulus, and Poisson's ratio of particles, respectively (Li, Xu, and Thornton 2005).

The input power is supplied by an electric motor and can be calculated from $P = T\omega$, where T and ω are the torque applied to the auger by seeds, and the auger angular speed, respectively (Beer et al. 2014).

2.2. Taguchi algorithm and analysis of variance

Taguchi was a Japanese engineer who designed a method in 1940s to balance quality, cost and variety of products. In the Taguchi method, the effective parameters in an experiment are called factors and the number of values assigned to each factor is named levels. To determine the ranking of each factor, the minimum required number of simulations is calculated as follows (Taguchi and Jugulum 2002):

$$RN = (N_l)^{N_f}, \quad (10)$$

where N_f and N_l are the numbers of factors and levels, respectively. Using the orthogonal arrays obtained from Taguchi algorithm, the number of simulations can be largely reduced. In the present study, 4 factors and 4 levels for each factor are considered, resulting in $4^2 = 16$ by using the Taguchi method, instead of $4^4 = 256$ simulations to study the effects of all factors.

In statistical problems, analysis of variance (ANOVA) is a useful method including the following operations (Montgomery and Runger 2014):

1. Analyzing statistical models involving many factors and levels.
2. Comparing the mean squares using the F -test.
3. Using different tests to achieve a suitable statistical model.

In the analysis of variance, degrees of freedom of numerator and denominator and an index called F_0 are calculated to be used in the F -test. F_0 is then compared to the corresponding F -value obtained by assuming the data follow the F -distribution considering 95% confidence. If $F_0 > F$, the assumption is valid in the 95% confidence interval; otherwise, the assumption is wrong. The F table can be found in Appendix A. ANOVA is performed to analyze the

simulations results in the Minitab program. One of the important parameters in ANOVA is the confidence interval; $CI = 1 - \alpha$. α is the probability of failure of the considered assumption, being usually assumed to be 0.05, making the confidence interval 95% (Montgomery and Runger 2014).

2.3. Response surface method for optimization

The response surface method (RSM) is a combination of mathematical and statistical methods used to analyze and evaluate the response of a multi-parameter system. Assume that the function Y depends on two variables x_1 and x_2 , written as follows (Montgomery and Runger 2014):

$$Y = f(x_1, x_2) + \varepsilon, \quad (11a)$$

where ε represents the error observed in the response Y . If the expected response is denoted by $E(Y) = f(x_1, x_2)$, then the response surface is represented by:

$$\eta = f(x_1, x_2). \quad (11b)$$

A contour plot of the response includes the current operating conditions and specifically region of the optimum conditions, being the peak of the response surface. In this research, the response surface method is used to find the optimum state of the auger filler performance.

3. Simulation procedure

In this research, the flow of 20,000 black pepper seeds in an auger filler is modeled until the complete discharge of the hopper, and the effects of auger angular velocity, diameter, and pitch to diameter ratio, and the agitator angular velocity, on the mass flow rate and the motor required power are investigated. The mechanical properties of black pepper seeds and the steel auger filler parts, and the contact parameters of seeds and seeds-walls interactions are brought in Table 1. The values (levels) of the considered parameters (factors) are stated in Table 2. The angular speeds of the auger and the agitator are chosen from a semi-automatic auger filler catalogs of All-Fill Inc. In the present study, four types of auger with different diameters are selected for the simulations based on the standards of Auger Manufacturing Specialists Company (AMS) (www.augermfgspec.com), and four different ratios of pitch to diameter are chosen from Kase Conveyors Company manuals (www.kaseconveyors.com).

To simulate the pepper seeds motion in the auger filler, the following steps are performed:

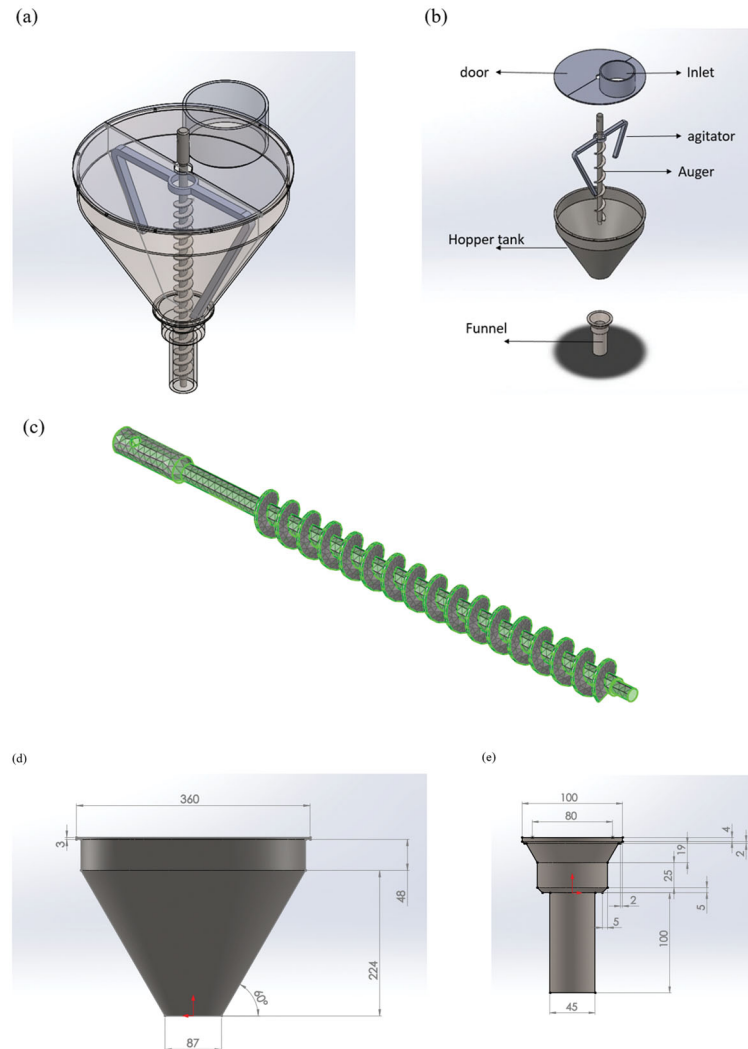
1. Geometric modeling of the auger filler in CAD software (SolidWorks).
2. Meshing of geometries and exporting the model as STL file (HyperView).
3. Importing the auger filler, seeds and other required data to LIGGGHTS software.
4. Simulating the problem through coding in LIGGGHTS, abbreviated for LAMMPS improved for general granular and granular heat transfer simulations.

Table 1. Material properties and contact parameters (Ghodki and Goswami 2017).

Material or collision parameters	Property	Value
Black pepper seeds	Density (ρ_p)	1,162 (kg/m ³)
	Shear modulus (G_p)	6.13×10^8 (Pa)
	Poisson's ratio (ν_p)	0.30
Stainless steel	Density (ρ_w)	7,800 (kg/m ³)
	Shear modulus (G_w)	7×10^{10} (Pa)
	Poisson's ratio (ν_w)	0.30
Seeds interaction	Coefficient of restitution (e_{pp})	0.10
	Coefficient of friction (μ_{s-pp})	0.45
	Coefficient of rolling resistance (μ_{r-pp})	0.01
Seeds-walls interaction	Coefficient of restitution (e_{pw})	0.20
	Coefficient of friction (μ_{s-pw})	0.45
	Coefficient of rolling resistance (μ_{r-pw})	0.01

Table 2. Different values of the considered design parameters.

	Auger angular velocity (rpm)	Agitator angular velocity (rpm)	Auger diameter (in.)	Auger pitch to diameter ratio (P/D)
1	250	25	1.25	0.5
2	400	35	1.5	0.66
3	550	45	1.75	1
4	700	55	2	1.5

**Figure 4.** (a) The auger filler model, (b) the exploded view, (c) view of the meshed auger, (d) dimensions of the tank in mm, and (e) dimensions of the funnel in mm.

5. Extracting the numerical results.
6. Post-processing and analyzing the data through statistical software (Minitab).

The simulated auger filler's isometric and exploded views, the meshed auger and dimensions of the tank and funnel are observed in Figure 4.

The pepper seeds flow is simulated in LIGGGHTS, Version 3.X (www.cfdem.com) and shown graphically in Paraview. The graphical view of the simulation environment is shown in Figure 5.

4. Results and discussion

One-way ANOVA is used to analyze the simulations results and to study the effect of the four parameters on both mass



Figure 5. A graphical view of the simulation.

flow rate and the required power. ANOVA general linear model is used to analyze Minitab. Then, the response surface optimizer of Minitab is utilized to maximize the mass flow rate and minimize the required power which is considered as the best outcome. The simulation details and response values for mass flow rate and torque applied to the auger, in each simulation are shown in Table 3.

4.1. Mass flow rate

The mass of exiting seeds varies linearly with time, as is illustrated in Figure 6. The mass flow rate is found to be constant by fitting a straight line on the discharged mass data with the correlation coefficient of approximate unity. In fact, the slope of the fitted line is used to determine the mass flow rate of discharge in all studied cases. It is observed in Figure 6 that the discharged mass becomes constant after 160 s showing that all the seeds left the filler.

The mass flow rate in the simulations with different auger angular velocities, diameters, pitch to diameter ratios and agitator angular velocity with 95% confidence intervals are shown in Figure 7. The mass flow rate is seen to increase by increasing the auger angular speed due to faster movements of the seeds. On the other hand, by increasing the agitator angular speed, the agitator conveys the seeds as it rotates and make them move in the horizontal planes, preventing the seeds from descending. This results in a slight decrease in mass flow rate as the agitator rotates faster. Indeed, as the seeds reach the funnel, they have no longer in contact with the agitator making the effect of agitator angular speed insignificant on the mass flow rate.

The seeds mass flow rate increases in cases of larger auger diameters, since as a consequence, the funnel diameter increases, and more seeds could leave the filler. The mass flow rate increases in the simulations with larger pitch-to-diameter ratios, due to more space between the auger threads and hence, the ability of seeds to exit the filler more easily.

Through ANOVA performed in Minitab, the condition of the maximum mass flow rate of seeds discharge is shown in Table 4.

Table 3. The simulation details and response values for mass flow rate and torque applied to the auger based on Taguchi algorithm.

Input and output quantities Simulation number	Auger angular velocity (rpm)	Agitator angular velocity (rpm)	Diameter of the auger (in.)	Pitch to diameter ratio	Mass flow rate (g/s)	Torque applied to the auger (N m)	Minimum power required (W)
1	250	25	1.25	0.5	9.49	37.58	983.8
2	250	35	1.50	1	61.7	23.66	619.4
3	250	45	1.75	0.66	60.9	26.72	699.5
4	250	55	2	1.50	23.5	16.51	432.2
5	400	25	1.5	0.66	51.0	30.92	1295
6	400	35	1.25	1.50	94.3	16.38	686.1
7	400	45	2	0.50	90.0	22.53	943.7
8	400	55	1.75	1	146	10.54	441.5
9	550	25	1.75	1.50	179	10.47	603.0
10	550	35	2	0.66	200	14.93	859.9
11	550	45	1.25	1	82.8	7.340	422.8
12	550	55	1.5	0.50	50.6	49.78	2867
13	700	25	2	1	335	12.14	889.9
14	700	35	1.75	0.50	99.3	34.08	2498
15	700	45	1.5	1.50	207	14.39	1,055
16	700	55	1.25	0.66	56.8	22.12	1,621

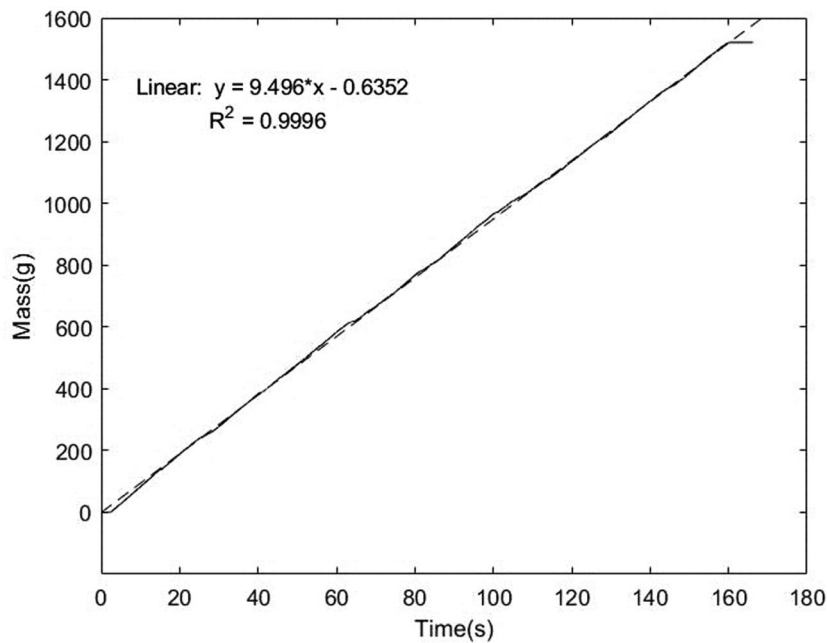


Figure 6. Variation of the pepper seeds mass flow rate with time (solid line) and the straight line fitted on the data (dashed line).

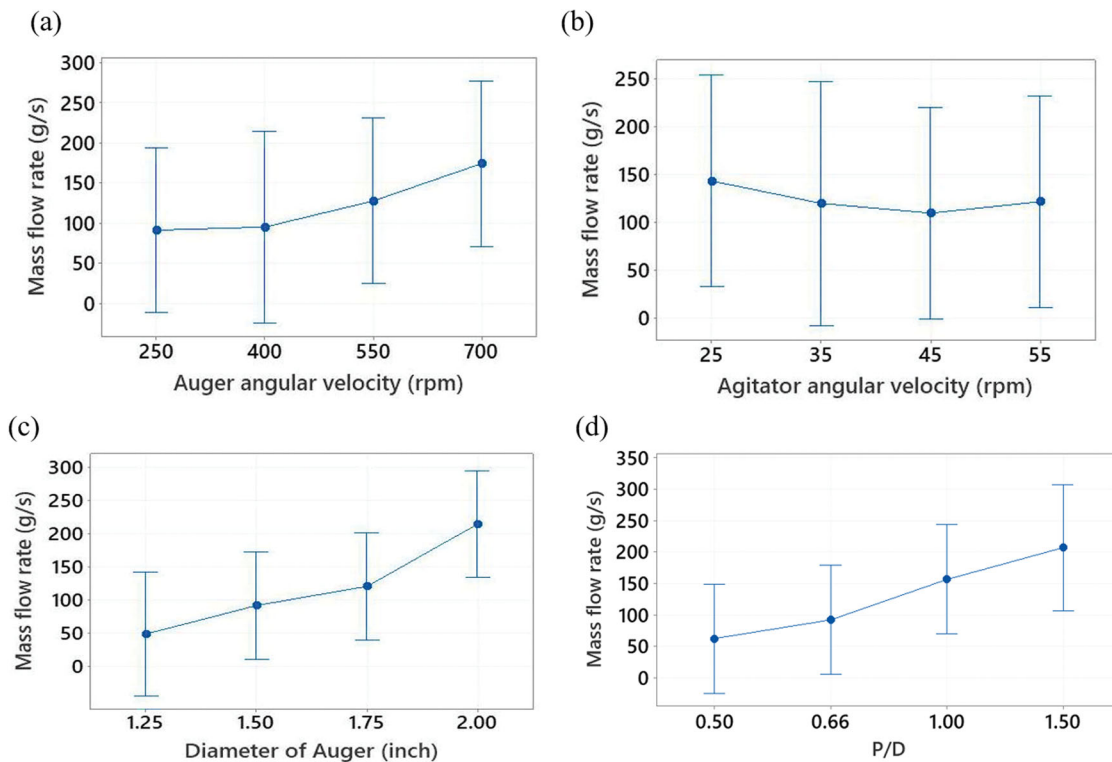


Figure 7. The mass flow rate of pepper seeds changed with different amounts of parameters with 95% confidence interval: (a) auger angular velocities, (b) agitator angular velocities, (c) auger diameters, (d) auger pitch-to-diameter ratios.

Table 4. Values of the four parameters resulted in reaching the maximum mass flow rate.

Auger angular velocity (rpm)	Agitator angular velocity (rpm)	Auger diameter (in.)	Auger pitch to diameter ratio (P/D)	Maximum mass flow rate (g/s)
700	25	2	1.5	340

4.2. The maximum torque and minimum required power

The torque applied to the auger by seeds repeated impacts is observed in Figure 8. The torque fluctuates a lot because of irregular collisions of seeds to the auger. After all, the motor

must still overcome the maximum torque applied to the auger. Multiplying the maximum torque and the auger angular velocity, the minimum power required to rotate the auger can be calculated in all simulations. The zero torques

observed after 160 s are attributed to the complete exit of seeds.

Figure 9 illustrates that the required power increases by increasing the auger angular velocity because of the direct proportionality of the auger speed and the required power. It can be observed in Figure 9, that the required power has no regular variation with respect to the agitator rotational speed and the auger diameter. As the auger pitch to diameter ratio decreases, the number of threads increases in a specific length of the auger, therefore, there happen more

contacts between the seeds and the auger threads, applying larger torque and consequently higher required power (Figure 9(d)).

Using ANOVA, the condition of minimum required power driving the auger can be seen in Table 5.

As it is observed in Tables 4 and 5, to have the maximum possible mass flow rate, the auger should rotate at its maximum speed being 700 rpm, while to have the minimum possible required power, its rotational speed should be decreased to 550 rpm. The agitator rotational speed is the same in both analyses which means it has no serious effect on the mass flow rate or the required power. Auger diameter shows the same aspect as the agitator rotational speed. The auger pitch to diameter ratio should be 1.5 to achieve the maximum mass flow rate while it should be 1 to have the minimum required power. Based on the results mentioned in Tables 4 and 5, it could be understood that the auger rotational speed and pitch-to-diameter ratio are more effective than the other studied parameters on the mass flow rate and the required power.

The best outcome among the completed simulations happens when the maximum mass flow rate occurs simultaneously with the minimum required power. Table 6 includes values of the four parameters to achieve this condition. Considering the jumps in the torque diagrams, a factor of safety of 2 was selected to ensure the reliable performance of the auger filler. Therefore, the motor power should be in the range of 800–1,000 W. The motor LS-XML-SX08ADKE belonged to the LS Company, South Korea (www.lsmecapion.com) with 800 W power is proposed to provide the required input power. The

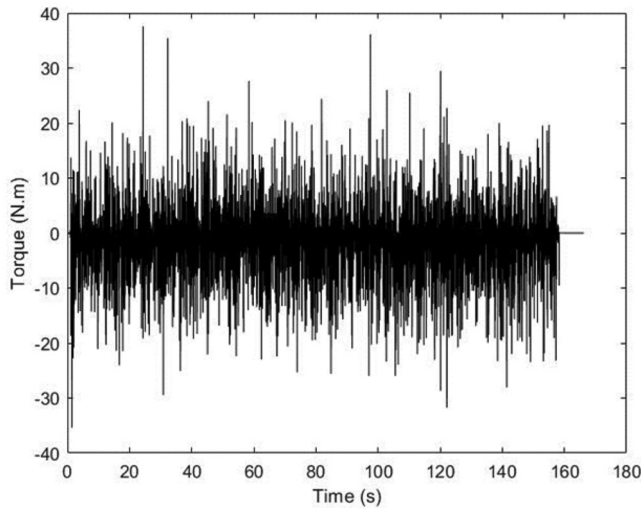


Figure 8. Time history of the torque applied to the auger by the pepper seeds.

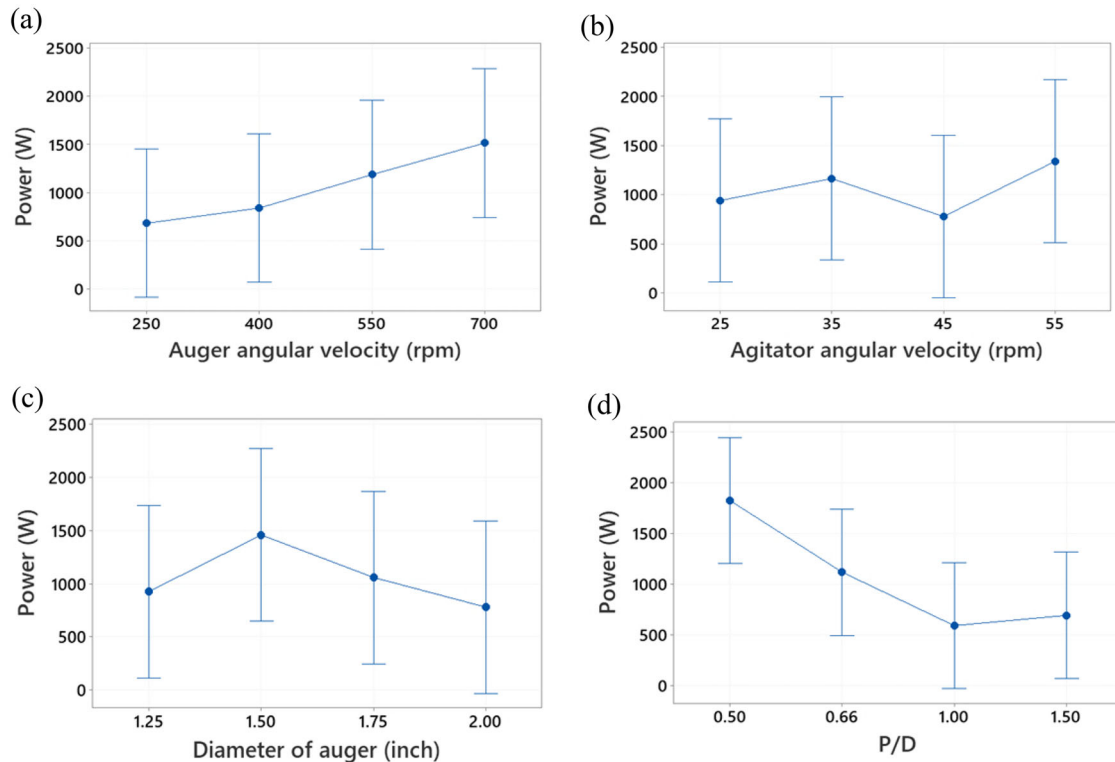


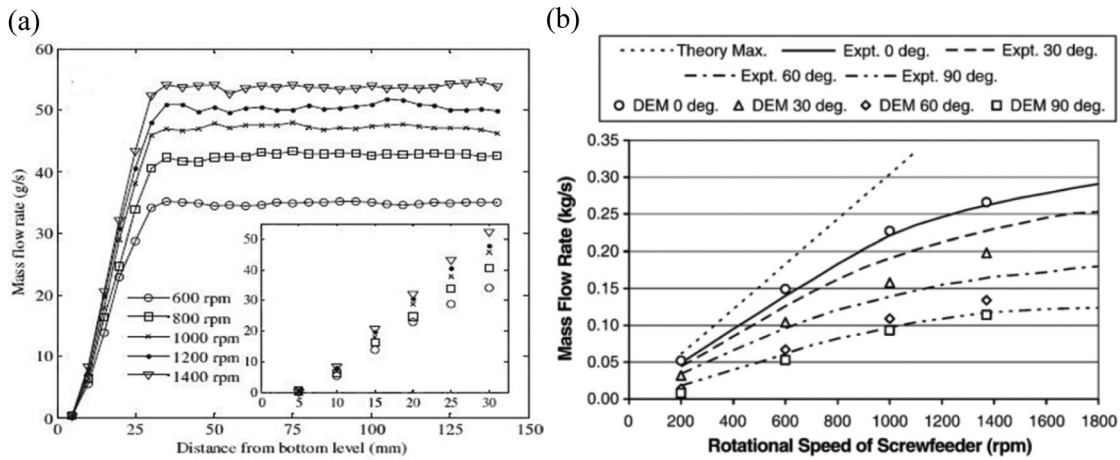
Figure 9. The required power of the motor driving the auger varied by different amounts of parameters with 95% confidence interval: (a) auger angular velocities, (b) agitator angular velocities, (c) auger diameters, (d) auger pitch-to-diameter ratios.

Table 5. Values of the four parameters resulted in reaching the minimum power.

Auger angular velocity (rpm)	Agitator angular velocity (rpm)	Auger diameter (in.)	Auger pitch to diameter ratio (P/D)	Minimum required power (W)
550	25	2	1	334

Table 6. Values of the four parameters resulted in reaching the maximum mass flow rate and minimum power.

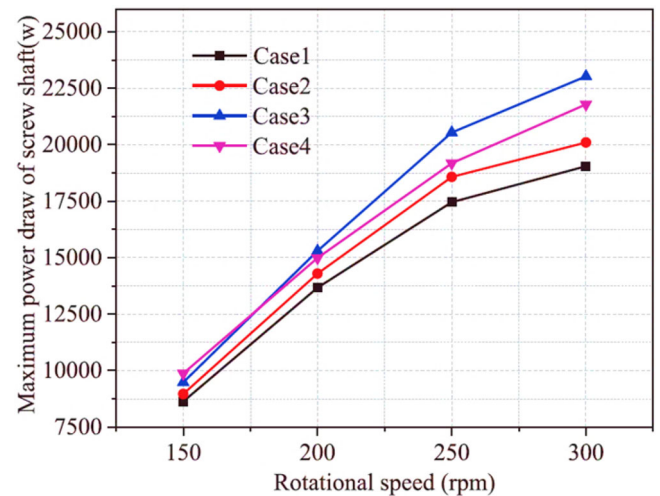
Auger angular velocity (rpm)	Agitator angular velocity (rpm)	Auger diameter (in.)	Auger pitch to diameter ratio (P/D)	Maximum mass flow rate (g/s)	Minimum required power (W)
550	25	2	1.5	300	434

**Figure 10.** (a) Mass flow rate of rice grains in a vertical rice mill, changed by the screw angular speed and the distance from the bottom surface (Han et al. 2017), (b) Variations of mass flow rate with the rotational speed and angle of inclination of the screw (Owen and Cleary 2009).

maintenance of the filler is very important, e.g., to avoid clogging of the seeds, frequent cleaning of the hopper and funnel is suggested. It is also advised that the motor is checked weekly for its accurate performance. The motor angular speed depends on the filling conditions and can be controlled accordingly.

4.3. Validation and comparison

According to Han et al. (2017), the mass flow rate of rice grains in a vertical rice mill increased by increasing the screw angular speed (Figure 10(a)) (Han et al. 2017). According to the experimental and DEM study of Owen and Cleary (2009), the mass flow rate of pellet particles in a screw conveyor increases by increasing the screw rotational speed (Figure 10(b)), which is similar to the results of the present research. Owen and Cleary (2009) also studied effect of the angle of inclination of the screw on the particles mass flow rate. They found that by increasing the angle of inclination of the screw, the mass flow rate of the particles decreases. The mass flow rate also compared in different angle of inclination of the auger the results show the mass flow rate decreased by increasing the angle of inclination. Yuan et al. (2020) found that the power required to rotate the screw transporting soybean seeds increased by increasing the screw angular speed (Figure 11) (Yuan et al. 2020). While the screw diameter increases from case 1 to case 4 in Figure 11, screw in case 3 is observed to consume larger power comparing to the other cases. Therefore, they

**Figure 11.** Power required to rotate the screw transporting soybean seeds varied by the screw diameter and angular speed (Yuan et al. 2020).

concluded that the power showed no regular variation with respect to the screw diameter. As found in the current research the results of the present study are qualitatively similar to the two articles mentioned above; however, due to different studied grains and problem dimensions, the results cannot be compared quantitatively. As another comparison, it should be noted that other parameters like humidity, temperature and hardness can all the flow rate so affect the mass flow rate or the required power. These parameters can change which is always intended to be constant without any

fluctuations to provide accuracy in packaging. For example, humidity would result in the particles cohesion and clogging. High temperature makes the seeds more ductile and changes their stiffness and contact properties, especially their coefficient of restitution. Indeed, for better performance of the device, it is recommended that auger filler works in a dry environment and room temperature. Such parameters can be studied with CFD-DEM modeling in future studies.

5. Conclusions

The main purpose of this study was to investigate the effects of 4 parameters including auger angular velocity, diameter and pitch to diameter ratio and the agitator angular velocity on the mass flow rate of black pepper seeds leaving an auger filler and the motor required power. The results showed that the seeds mass flow rate increases by increasing the auger angular velocity, diameter and pitch to diameter ratio, while slightly decreases by increasing the agitator angular velocity. The auger diameter was found to be the most important parameter affecting the mass flow rate. The motor required power increases by increasing the auger angular velocity and decreases as the auger pitch-to-diameter ratio increases. However, the required power shows no regular variation with the auger diameter and the agitator angular velocity. The best outcome from simulations is achieved when the maximum mass flow rate happens with minimum power. The maximum mass flow rate of 0.3 kg/s occurs with the minimum power of 434 W when the auger and agitator angular speeds are 550 rpm and 25 rpm, respectively, and the auger diameter and pitch to diameter ratio were 2 in. and 1.5, respectively.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figure A1. The F table used in ANOVA method (Montgomery and Runger 2014).